

Responsive Student Registration and Result Management System

HTML5, CSS3 and JavaScript

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Programme: MCA 2026-28

Introduction

This assignment is a Responsive Student Registration and Result Management System built with HTML5, CSS3 and JavaScript. A student enters their details and marks in a form, and the page then works out the result on its own, without any server.

The program follows this flow:

- The student enters name, email, phone, course and marks in HTML, CSS and JavaScript.
- JavaScript validates the data (name, email, phone, course, and marks between 0 and 100).
- It calculates the total (out of 300) and the percentage.
- It decides Pass or Fail. A student passes only when every subject has at least 40 marks.
- The result is displayed dynamically on the page, in green for Pass and red for Fail.
- The student object is converted to JSON with `JSON.stringify()`.
- The JSON is stored in the browser's Local Storage under the key student.

My Approach

I built the project in two stages instead of writing everything at once.

Stage 1: HTML and JavaScript only. I first wrote just the HTML form and the script, with no CSS. I opened the file in the browser and tested the logic: the validation messages, the total and percentage, the Pass/Fail result, and the data saved in Local Storage. I fixed problems at this stage, so the working logic was ready before any styling.

Stage 2: CSS with the browser's developer tools. After the logic worked, I did the styling in the browser itself. I opened the console tools (Inspect) and tried colours, spacing, fonts, the header layout and the responsive behaviour live on the page, and saw the changes straight away. Once the page looked right, I copied the final CSS from the tools and pasted it back into the code file, inside a `<style>` tag.

For responsiveness, the inputs use `width: 100%` and a media query for screens up to 600px reduces the spacing, so the same page fits both desktop and mobile widths.

The complete code is on the next pages, followed by the output screenshots.

Source Code

File: student_portal.html (HTML, CSS and JavaScript in one file)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1">
  <title>Student Result Management</title>
  <style>
    body {
      font-family: 'Times New Roman', Times, serif;
      background: #f3f4f8;
      margin: 0;
    }

    header {
      background: #a127ff;
      color: white;
      padding: 15px 20px;
      display: flex;
      justify-content: space-between;
      align-items: center;
    }

    header h1 {
      margin: 0;
    }

    nav a {
      color: white;
      margin-left: 15px;
      text-decoration: none;
      font-weight: bold;
    }

    section {
      background: white;
      max-width: 600px;
      margin: 20px auto;
      padding: 20px;
    }

    label {
      display: block;
      font-weight: bold;
      margin-bottom: 5px;
    }

    input, select {
      width: 100%;
      padding: 8px;
      box-sizing: border-box;
    }

    button {
      background: #a127ff;
      color: white;
```

```

    border: none;
    padding: 10px 20px;
    cursor: pointer;
}

#resetBtn {
    background: gray;
}

.pass { color: green; font-weight: bold; }
.fail { color: red; font-weight: bold; }

@media (max-width: 600px) {
    section {
        margin: 10px;
        padding: 15px;
    }
}
</style>
</head>
<body>

    <header>
        <h1>Student Portal</h1>
        <nav>
            <a href="#registration">Registration</a>
            <a href="#result">Result</a>
        </nav>
    </header>

    <section id="registration">
        <h2>Student Registration</h2>
        <form id="studentForm">
            <p>
                <label for="name">Student Name</label>
                <input type="text" id="name" placeholder="Enter name">
            </p>
            <p>
                <label for="email">Email</label>
                <input type="email" id="email" placeholder="Enter email">
            </p>
            <p>
                <label for="phone">Phone</label>
                <input type="text" id="phone" placeholder="10 digit phone
number" maxlength="10">
            </p>
            <p>
                <label for="course">Course</label>
                <select id="course">
                    <option value="" selected disabled>Choose your
course</option>
                    <option>MCA</option>
                    <option>MBA</option>
                    <option>BBA</option>
                    <option>B Tech CSE</option>
                </select>
            </p>
            <p>
                <label for="html">HTML Marks</label>
                <input type="number" id="html" placeholder="0 - 100">
            </p>
        </form>
    </section>

```

```

    <p>
      <label for="css">CSS Marks</label>
      <input type="number" id="css" placeholder="0 - 100">
    </p>
    <p>
      <label for="js">JavaScript Marks</label>
      <input type="number" id="js" placeholder="0 - 100">
    </p>
    <button type="submit">Generate Result</button>
    <button type="button" id="resetBtn">Reset</button>
  </form>
</section>

<section id="result">
  <h2>Result</h2>
  <div id="output"><p>No result generated yet.</p></div>
</section>

<script>
  document.getElementById("studentForm").addEventListener("submit",
function (e) {
  e.preventDefault();

  // 1. Read the values
  var name = document.getElementById("name").value.trim();
  var email = document.getElementById("email").value.trim();
  var phone = document.getElementById("phone").value.trim();
  var course = document.getElementById("course").value;
  var html = document.getElementById("html").value;
  var css = document.getElementById("css").value;
  var js = document.getElementById("js").value;

  // 2. Validate
  if (!/^[A-Za-z ]+$/.test(name)) {
    alert("Name should contain only letters");
    return;
  }
  if (!/^[A-Za-z0-9._-]+@[A-Za-z0-9-]+\.[A-Za-z.]+$/.test(email)) {
    alert("Enter a valid email");
    return;
  }
  if (!/^[0-9]{10}$/.test(phone)) {
    alert("Phone number must be 10 digits");
    return;
  }

  if (course === "") {
    alert("Please choose your course");
    return;
  }

  if (html === "" || css === "" || js === "") {
    alert("Enter all marks");
    return;
  }

  html = Number(html);
  css = Number(css);
  js = Number(js);

```

```

        if (html < 0 || html > 100 || css < 0 || css > 100 || js < 0 ||
js > 100) {
            alert("Marks must be between 0 and 100");
            return;
        }

        // 3. Calculate
        var total = html + css + js;
        var percentage = (total / 300) * 100;
        var result = "Pass";
        if (html < 40 || css < 40 || js < 40) {
            result = "Fail";
        }

        // 4. Display the result
        document.getElementById("output").innerHTML =
            "<p><b>Name:</b> " + name + "</p>" +
            "<p><b>Email:</b> " + email + "</p>" +
            "<p><b>Phone:</b> " + phone + "</p>" +
            "<p><b>Course:</b> " + course + "</p>" +
            "<p><b>HTML:</b> " + html + " | <b>CSS:</b> " + css + " |
<b>JavaScript:</b> " + js + "</p>" +
            "<p><b>Total:</b> " + total + " / 300</p>" +
            "<p><b>Percentage:</b> " + percentage.toFixed(2) + "%</p>" +
            "<p class='" + result.toLowerCase() + "'><b>Result:</b> " +
result + "</p>";

        // 5. Convert to JSON and store in Local Storage
        var student = {
            name: name,
            email: email,
            phone: phone,
            course: course,
            marks: { html: html, css: css, js: js },
            total: total,
            percentage: percentage.toFixed(2),
            result: result
        };

        localStorage.setItem("student", JSON.stringify(student));
    });

    document.getElementById("resetBtn").addEventListener("click",
function () {
        document.getElementById("studentForm").reset();
        document.getElementById("output").innerHTML = "<p>No result
generated yet.</p>";
    });
</script>

</body>
</html>

```

Output

The screenshot shows a web browser window with the address `file:///Users/deeyamathur/fullstack/student_portal.html`. The page has a purple header with "Student Portal" on the left and "Registration Result" on the right. The main content area is divided into two sections: "Student Registration" and "Result".

Student Registration

Student Name: Deeya Mathur
Email: mca2026.deeyam@isu.ac.in
Phone: 7006021614
Course: MCA
HTML Marks: 100
CSS Marks: 95
JavaScript Marks: 92
Buttons: Generate Result, Reset

Result

Name: Deeya Mathur
Email: mca2026.deeyam@isu.ac.in
Phone: 7006021614
Course: MCA
HTML: 100 | CSS: 95 | JavaScript: 92
Total: 287 / 300
Percentage: 95.67%
Result: Pass

Figure 1: Desktop view. Valid details with marks 100, 95 and 92 give a total of 287 / 300, 95.67%, and the result is Pass.

The screenshot shows the mobile-width view of the Student Portal. The layout is a single column. The header is purple with "Student Portal" on the left and "Registration Result" on the right. The main content area is divided into two sections: "Student Registration" and "Result".

Student Registration

Student Name: Deeya Mathur
Email: mca2026.deeyam@isu.ac.in
Phone: 7006021614
Course: MCA
HTML Marks: 100
CSS Marks: 95
JavaScript Marks: 92
Buttons: Generate Result, Reset

Result

Name: Deeya Mathur
Email: mca2026.deeyam@isu.ac.in
Phone: 7006021614
Course: MCA
HTML: 100 | CSS: 95 | JavaScript: 92
Total: 287 / 300
Percentage: 95.67%
Result: Pass

Figure 2: Mobile-width view of the same result. The layout stays in a single column and fits the narrow screen.

The screenshot shows the mobile-width view of the Student Portal with a failed result. The layout is a single column. The header is purple with "Student Portal" on the left and "Registration Result" on the right. The main content area is divided into two sections: "Student Registration" and "Result".

Student Registration

Student Name: Deeya Mathur
Email: mca2026.deeyam@isu.ac.in
Phone: 7006021614
Course: MCA
HTML Marks: 100
CSS Marks: 20
JavaScript Marks: 92
Buttons: Generate Result, Reset

Result

Name: Deeya Mathur
Email: mca2026.deeyam@isu.ac.in
Phone: 7006021614
Course: MCA
HTML: 100 | CSS: 20 | JavaScript: 92
Total: 212 / 300
Percentage: 70.67%
Result: Fail

Figure 3: Fail case. CSS marks are 20, below the 40 pass mark. Total 212 / 300, 70.67%, result Fail (in red).

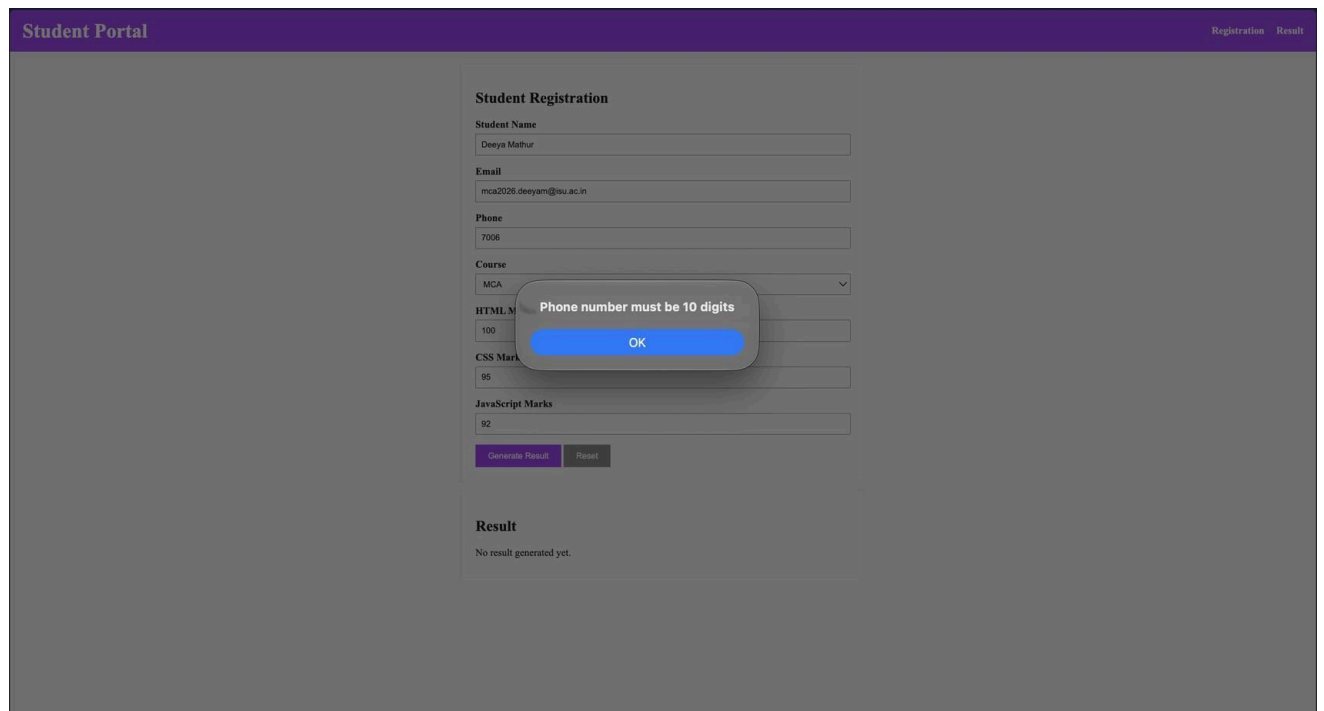


Figure 4: Validation alert. A phone number with only 4 digits shows "Phone number must be 10 digits", and no result is generated.

Student Registration

Student Name

Email

Phone

Course

HTML Marks

CSS Marks

JavaScript Marks

Result

Name: Deeya Mathur

Email: mca2026.deeyam@isu.ac.in

Phone: 7006021614

Course: MCA

HTML: 100 | **CSS:** 20 | **JavaScript:** 92

Total: 212 / 300

Percentage: 70.67%

Result: Fail

Figure 5: Email validation. An email without '@' is rejected by the email field and the browser shows a message.

Student Portal

RegistrationResult

Student Registration

Student Name
Deeya Mathur

Email
mca2026.deeyam@isu.ac.in

Phone
7006201614

Course
MCA

HTML Marks
95

CSS Marks
30

JavaScript Marks
99

Generate ResultReset

Result

Name: Deeya Mathur
Email: mca2026.deeyam@isu.ac.in
Phone: 7006201614
Course: MCA
HTML: 95 | CSS: 30 | JavaScript: 99
Total: 224 / 300
Percentage: 74.67%
Result: Fail

Figure 6: Another Fail case. HTML 95, CSS 30, JavaScript 99 give 224 / 300 and 74.67%, but the result is Fail because CSS is below 40.

Student Portal

RegistrationResult

Student Registration

Student Name
Deeya Mathur

Email
mca2026.deeyam@isu.ac.in

Phone
7006201614

Course
MCA

HTML Marks
100

CSS Marks
95

JavaScript Marks
92

Generate ResultReset

Result

Application

file:///

Key

Value

student

{ "name": "Deeya Mathur", "email": "mca2026.deeyam@isu.ac...."

Storage

Local storage

file:///

Session storage

Extension stora...

IndexedDB

Cookies

Private state to...

Interest groups

Shared storage

Cache storage

Storage buckets

Background services

Back/forward c...

Background fet...

Background sync

Bounce trackin...

Notifications

Payment handler

Periodic backg...

Speculative loa...

Push messaging

Reporting API

Frames

top

{ "name": "Deeya Mathur", "email": "mca2026.deeyam@isu.ac.in", "phone": "7006201614", "course": "MCA", ... }

course: "MCA"

email: "mca2026.deeyam@isu.ac.in"

marks: {html: 100, css: 95, js: 92}

name: "Deeya Mathur"

percentage: "95.67"

phones: "7006201614"

result: "Pass"

total: 287

Figure 7: Local Storage in the browser's developer tools (Application tab). The key student holds the student object as JSON, with name, email, phone, course, marks, total, percentage and result.